## Combinations

Name _____

Pick 3 students from 5 (A, B, C, D, E) to be on a committee. All 3 have the same job. Choosing A, B, C is the **same committee** as choosing C, A, B. **Order does not matter.**

Count as if order mattered: ${}_5P_3 = 5 \cdot 4 \cdot 3 = 60$
Each committee of 3 was counted 3! = 6 times: ABC, ACB, BAC, BCA, CAB, CBA
Divide: $60 \div 6 =$ **10** committees

**Combination formula:** ${}_nC_r = \binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$ choose $r$ things from $n$ things

${}_nC_r$ and $\binom{n}{r}$ are both read "$n$ choose $r$." Some books write $C(n, r)$.

**Example:** ${}_5C_3 = \dfrac{5!}{3!\,2!} = \dfrac{120}{6 \cdot 2} = 10$

**Permutation or combination?** Ask: *If I switch the order, is it a different result?*

| **Order matters: permutation** | **Order does not matter: combination** |
|---|---|
| President, vice president, secretary (different jobs) | A committee of 3 (same job) |
| 1st, 2nd, 3rd place | Picking 3 winners of the same prize |
| Arranging books on a shelf | Choosing 3 books to take on a trip |

**Handshakes.** 10 people each shake hands once with everyone else. A handshake is a pair, and Ana-Ben is the same as Ben-Ana: ${}_{10}C_2 = \dfrac{10 \cdot 9}{2} = 45$ handshakes.

Work with a partner. Compare answers when you finish each problem.

1. Find each value.

   (a) ${}_5C_2$

   (b) ${}_6C_3$

   (c) ${}_8C_1$

   (d) $\binom{7}{7}$

2. Permutation or combination? Then find the number of ways.

   (a) Choose 3 pizza toppings from 10.

   (b) Give 1st, 2nd, and 3rd prizes to 3 of 10 people.

   (c) Choose a president and a treasurer from 12 people.

   (d) Choose 4 people from 15 for a committee.

3. 12 people meet. Each person shakes hands once with every other person. How many handshakes?

4. A restaurant has 8 toppings. A pizza has 2 different toppings. How many different 2-topping pizzas are there?

5. Compare ${}_{10}P_3 = 720$ and ${}_{10}C_3 = 120$.

   (a) Why is ${}_{10}P_3$ bigger?

   (b) What number do you divide by to get from 720 to 120? Why?

6. A hand of 5 cards is dealt from a deck of 52 cards. Use a calculator. How many different hands are there?

7. A student says, “Choosing 2 of 6 students to clean the board is a permutation, because there are 2 people.” Do you agree? Explain.